Krishna Gupta

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PROFESSIONAL SUMMARY

Innovative Software & AI Engineer and B.Tech in Data Science student (Class of 2027) ranking in the **97th National Percentile (611/800, 100% in Coding, 87% in Computer Fundamentals) in the Aon CoCubes National Employability Assessment** with verified High Employability for Software Engineering roles. Progressive industry internship experience architecting scalable distributed systems, production AI pipelines, and high-throughput data microservices. **Possesses a solid foundation in core system design and software architecture principles** (High-Level/Low-Level Design, microservices vs. monoliths, distributed caching hierarchies, load balancing, database sharding & indexing, CAP theorem, and event-driven patterns). Proven expertise across the MLflow production lifecycle (tracing, evaluation, prompt templates, AI gateways, agent servers), full-stack microservices with 99.9% uptime, and certified AWS Developer skills in FastAPI, PyTorch, and relational/NoSQL optimization.

EDUCATION & STANDARDIZED ASSESSMENT

Oriental Institute of Science and Technology Bhopal, Madhya Pradesh, India
Bachelor of Technology (B.Tech) in Data Science (Class of 2027) Aug 2023 – May 2027
• **Aon CoCubes National Employability Assessment (Jun 2026):** Overall Score **611/800** ("Excellent") | **97th National Percentile** • 98th Regional Percentile | **100% Score in Coding (5/5) • 87% in Computer Fundamentals (5/5)** | Certified High Job Suitability for **Software Developer & Software Engineer** roles.

TECHNICAL SKILLS

System Design & Architecture: Core System Design Principles, High-Level Design (HLD), Low-Level Design (LLD), Client-Server & Microservices Architecture, Monolith vs. Distributed Systems, Scalability & High Availability, Load Balancing (Round-Robin, Least Connection), Reverse Proxies & API Gateways (Nginx), Distributed Caching Strategies (Redis, In-Memory, Cache-Aside), Database Indexing, Partitioning & Sharding, CAP Theorem & ACID Transactions, Asynchronous & Event-Driven Architecture, Message Queues, Rate Limiting, Fault Tolerance, Resiliency & Graceful Degradation

Generative AI & Agentic Systems: Large Language Models (LLMs), Vision-Language Models (VLMs), LangGraph, LangChain, Autonomous Multi-Agents, Hybrid Retrieval-Augmented Generation (Hybrid RAG), Prompt Engineering, Safety Guardrails, Model Alignment (RLHF/DPO), Vector Databases (FAISS, pgvector), Hugging Face Transformers

MLOps & Cloud Infrastructure: MLflow (Tracing, Model Evaluation, Prompt Templates, AI Gateways, Agent Server, Hyperparameter Tuning), Experiment Tracking, Model Registry, CI/CD Pipelines, Docker Containers, AWS Cloud Infrastructure (EC2, S3, IAM, CloudWatch)

Backend, APIs & Databases: Python (Advanced), Java, SQL (PostgreSQL, MySQL), FastAPI, RESTful APIs, Microservices, Node.js, React.js, ETL/ELT Pipelines, Dimensional Modeling (Star/Snowflake Schemas), Git, Linux Server Administration, Agile/SDLC

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Classification, Regression, Time-Series Forecasting (LSTM), Quantitative Financial Modeling, Computer Vision (OpenCV, EasyOCR), Deep Neural Networks (CNNs), PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

WORK EXPERIENCE

NexoraMind Tech Remote
AI Engineer Intern (6-Month Program) Sep 2026 – Present
- Architecting and developing production-grade AI microservices, intelligent agentic workflows, and automated pipeline solutions leveraging modern LLMs and deep learning frameworks.
- Building scalable data processing and vector-based semantic retrieval engines (RAG), optimizing inference latency, context window utilization, and query precision.
- Formulating robust evaluation benchmarks, capability matrices, and safety guardrails to ensure end-to-end AI system reliability and enterprise compliance.

Ethara.AI Remote
LLM & AI Post Training Intern (Paid Internship) Feb 2026 – May 2026
- Benchmarked and evaluated 50,000+ Large Language Model outputs within LLMOps workflows, boosting model alignment, precision, and safety guardrails by 15% across diverse prompt distributions.
- Designed and automated scalable Python data preprocessing, feature extraction, and validation pipelines, saving 20+ hours per week in manual QA and dataset screening.
- Formulated structured capability rubrics and collaborated with AI research teams to analyze edge cases and streamline dataset curation for reinforcement learning environments.

Kanchan Pvt Ltd – Web Development Wing (Sapphire) Remote
Full Stack Development Intern (Paid Internship) Oct 2025 – Feb 2026
- Understood and applied core system design principles to architect and deploy full-stack web infrastructure for review analytics platform "RevU Social" (revu.social) using React.js and Node.js REST APIs.
- Optimized PostgreSQL and MySQL relational database schemas, query execution plans, and indexing strategies, reducing query latency and accelerating transaction speeds by 25%.
- Implemented secure JWT authentication protocols, API rate limiting, and deployed Docker containerized microservices on Linux cloud servers, maintaining 99.9% uptime.

KEY TECHNICAL PROJECTS

Stock Market Future – AI-Powered Financial Forecasting & Analytics Platform 2024 – Present

Technologies: Python, FastAPI, PyTorch (LSTM + Transformers), Scikit-learn, MLflow, React 19, Tailwind, Lightweight Charts, Firebase Auth

Live App: smf-ai.web.app | GitHub: github.com/KGupta171025/Stock-Market-Future

- Engineered an AI financial prediction platform integrating hybrid ensemble models (LSTM + Transformers) and Scikit-learn to forecast asset trends with 88% directional precision.
- Architected real-time technical charting (EMA, RSI, MACD), financial news sentiment analysis, and AMFI mutual funds analytics with MLflow hyperparameter tuning.
- Implemented automated risk assessment workflows calculating entry zones, target price projections, and stop-loss levels for multi-timeframe financial analysis.
- Engineered production deployments on Firebase Auth and Cloudflare Workers with client-side caching and streaming data endpoints, achieving sub-100ms response times.

KALKI 1.5 – Autonomous Multi-Agent OS & MLOps Platform 2024 – Present

Technologies: Python, FastAPI, LangGraph, LangChain, LLMs, VLMs, MLflow Tracing/Evaluation, Hybrid RAG, FAISS

Live Demo: kalki.hg497kg.workers.dev | GitHub: github.com/KGupta171025/KALKI-1.5

- Engineered an autonomous multi-agent operating system integrating LLMs and Vision-Language Models (VLMs) via LangGraph and LangChain to automate complex multi-step reasoning workflows with decoupled agent boundaries.
- Architected a low-latency Hybrid RAG pipeline integrating dense vector embeddings and FAISS with in-memory caching mechanisms, cutting semantic search query retrieval latency by 40%.
- Integrated MLflow tracing, prompt evaluation templates, and automated guardrail logging to monitor inter-agent communication, audit token consumption, and ensure deterministic execution safety.

Score Vision - AI (FOG Tech) – Computer Vision & Document OCR Extraction Platform 2025 – Present

Technologies: Python, OpenCV, EasyOCR, Sentence-Transformers, FAISS, FastAPI, React, Docker

Live Demo: scorevision-ai.hg497kg.workers.dev | GitHub: github.com/KGupta171025/FOG_Technologies

- Developed an intelligent OCR and computer vision pipeline achieving 92% text extraction accuracy and sub-200ms semantic matching on complex, unstructured imagery.
- Designed a decoupled client-server microservice architecture using FastAPI and Docker to handle high-throughput image processing and FAISS vector similarity search at scale.

SHELF-SCANNER – High-Speed Computer Vision & Document OCR Platform 2025 – Present

Technologies: Python, OpenCV, EasyOCR, Sentence-Transformers, FAISS, FastAPI, Cloudflare Workers

Live Demo: shelf-scanner.hg497kg.workers.dev | GitHub: github.com/KGupta171025/SHELF-SCANNER

- Engineered an image-to-text OCR and semantic book-matching platform using EasyOCR and FAISS vector search, achieving sub-200ms query retrieval latency.
- Architected serverless edge worker routing on Cloudflare to process incoming visual payload streams with minimal infrastructure footprint.

NITI AI – Multilingual Scheme Intelligence & Financial Incentive Platform 2025 – Present

Technologies: Next.js 14, React 18, TypeScript, Tailwind CSS, Cloudflare Workers, Firebase Auth, Cloud Firestore

Live App: niti--ai.web.app | GitHub: github.com/KGupta171025/NITI-AI

- Architected a multilingual AI scheme intelligence platform evaluating demographic eligibility, enterprise sector, and funding needs against 100+ national and state schemes.
- Engineered serverless edge deployments on Cloudflare Workers and Firebase with static asset streaming and client-side caching, achieving sub-100ms response times.

RevU Social – Full-Stack Review & Analytics Platform Oct 2025 – Feb 2026

Technologies: React.js, Node.js, REST APIs, MySQL, PostgreSQL, Star Schema | *revu.social*

- Designed dimensional star-schema database models in MySQL/PostgreSQL and architected modular API endpoints to power real-time business intelligence and performance dashboards.
- Integrated asynchronous review aggregation pipelines with tokenized authentication and role-based access control.

Razorpay Recovery – Smart Payment Recovery & FinTech Transaction Pipeline 2025

Technologies: Python, SQL, PostgreSQL, REST APIs, Webhooks, Data Pipelines, Transaction Analytics

Live Demo: razor-recovery.hg497kg.workers.dev | GitHub: github.com/KGupta171025/Razor-Recovery

- Constructed an automated, event-driven transaction recovery pipeline capturing failed payment webhooks in real time with idempotent processing, successfully recovering 18% of abandoned checkouts.
- Designed analytical star-schema databases in PostgreSQL to aggregate transaction failure patterns, automate notification triggers, and generate recovery analytics.

CERTIFICATIONS & STANDARDIZED ASSESSMENT

- **Aon CoCubes Assessment:** 97th %ile (611/800, 100% Coding)
- **Machine Learning with Python** (IBM SkillsBuild, Jun 2026)
- **Data Science & Analytics** (HP LIFE, Jun 2026)
- **AWS Certified Developer Associate** (Infosys, Jun 2026)
- **Neural Networks & CNNs** (LinkedIn, Jun 2026)
- **Deloitte Australia Data Analytics Simulation** (Sep 2025)

RELEVANT COURSEWORK

System Design & Architecture • Distributed Systems • Data Structures & Algorithms • Database Management Systems • Operating Systems • Computer Networks • Software Engineering • Machine Learning • Big Data Engineering